

Disaster Risk Management

Plans and actions that are put in place to strengthen community resilience and manage disaster risks

Aim/Goal	Reducing			Increase preparedness for disaster response			Responses <i>(in the immediate hours and days)</i>			Recovery <i>(over a longer period of time)</i>
	Exposure	Vulnerability								
Strategy	Landuse planning	Hazard-resistant building designs	Monitoring and Warning System	Public awareness of hazards through education	First-aid training	Conducting Evacuation Drills	Search and Rescue	Timely Evacuation	Provision of basic social & psychological services	Restoration and improvement of facilities and living conditions
How will this strategy help to strengthen community resilience? Use examples to support	Land use planning reduces exposure by preventing development in high-risk areas, lowering loss of lives and property. For example, coastal communities in Japan were relocated to higher ground, reducing tsunami damage.	Hazard-resistant building designs reduce vulnerability by preventing building collapse during earthquakes. For example, Taipei 101’s mass damper reduces swaying and structural damage.	Monitoring and warning systems provide early alerts that allow people to take protective action or evacuate. For example, Japan’s early warning system stopped bullet trains during the 2011 Tohoku earthquake, saving lives.	Public awareness of hazards through education builds community resilience by teaching people how to respond correctly during earthquakes and volcanic eruptions, reducing injuries and panic. For example, people who understand earthquake early warnings know when to take cover or evacuate.	First-aid training strengthens community resilience by enabling people to provide immediate medical assistance to the injured, reducing deaths and allowing survivors to evacuate safely.	Conducting evacuation drills builds resilience by familiarising people with evacuation routes and safe areas, reducing the likelihood of being trapped. For example, during the 2011 Tohoku earthquake and tsunami, students in Kamaishi, Japan evacuated quickly to higher ground, saving nearly 3,000 lives.	Search and rescue strengthens community resilience by saving lives immediately after a disaster, allowing more people to contribute to recovery. For example, during the 2020 Aegean Sea earthquake in Turkey, over 100 survivors were rescued, reducing long-term social disruption.	Timely evacuation reduces loss of life from disasters such as aftershocks, tsunamis, or volcanic eruptions. In the 2011 Tohoku earthquake in Japan, students in Kamaishi city were evacuated to higher ground, saving nearly 3,000 students.	Providing basic social & psychological services help makes communities stronger by taking care of immediate needs and mental health. After the 2010 Haiti earthquake, clean water was given to stop diseases, and after the 2018 Palu earthquake, counselling helped people deal with trauma.	Restoration and improvement of facilities and living conditions build community resilience by rebuilding infrastructure to be more hazard-resistant, reducing damage and loss in future disasters.
What are its limitations?	People may still live or build in risky areas despite rules, and it cannot stop disasters from happening.	Expensive to build, and older buildings may not be upgraded.	Cannot prevent disasters, and warnings may be too short or fail if technology breaks down.	Effectiveness is limited by low literacy, cultural beliefs, or lack of funding.	Not everyone may have access or retain skills; emergencies can still overwhelm individuals.	People may forget procedures, panic during real events, or drills may not cover all scenarios.	Can be slow if there are not enough trained workers, equipment, or clear access to affected areas.	May fail if there is little warning, poor transport, or people do not follow instructions.	Can be difficult when there is not enough food, water, medicine, money, or helpers.	Takes a lot of time and money, and poorer countries may not be able to rebuild stronger buildings
Remarks	The more ready a country is in reducing exposure and vulnerability through the strategies here, the less damage and death sustained. This <u>reduces the strain placed on first-aid respondents to response to an overwhelmingly large area</u> . This allows <u>necessary responses to be more targeted and effective</u> . This would also <u>increase their ability to recover quickly and there is less restoration</u> or rebuilding that needs to be done.			These strategies are done and practiced before the disaster. They help with the efficiency of <u>responses during and after the disaster</u> . Without these preparedness measures, irrational behaviours and the lack of immediate help might occur, leading to more death.			This comes with <u>readiness to know how to react</u> . How effective this can be done is dependent on their readiness which is comes from increased preparedness through education, first-aid training and drills. To better response and recover, one must be more prepared and know what needs to be done. If the community can do that, the need to provide immediate responses will be less strenuous on the government.			<i>See reduction of exposure and vulnerability.</i>
General Challenges	Extent of community resources - This includes technological and financial resource. - Developing countries often lack the resources to build community resilience. - They may also choose to prioritise economic development instead of preparing for and dealing with tectonic disasters. - For example, Dhaka, Bangladesh is very vulnerable to earthquakes as 30% of its people live below the poverty line, with much of the country’s resources devoted to economic development instead of reinforcement of buildings						Lack of domestic resources This includes technological and financial resource. For example, during the 2010 Haiti mag 7.0 earthquake, there was a shortage of ambulances to transport the injured, and heavy machinery to remove the rubble from collapsed buildings. As a result, rescue teams spent many hours to remove the rubble manually which slowed down rescue efforts			
	The capability of the community to organize itself for disasters - This could be limited by lack of efforts to educate and train the community to respond and recover and/or political instability, and or corruption which may lead to loss of funding - For example, the community in Haiti was not able to organize itself for the 2010 mag 7.0 earthquake as there were no measure put in place to educate Haitians on what to do during a disaster. They also had no network of community healthcare workers which limited the medical and emergency responses. There was chaos and unrest as survivors found it hard to take care of themselves						Challenge in engaging relevant stakeholders to collaborate and integrate disaster management strategies into their practices. • Disagreements on strategies. • Stakeholder underestimating or lack awareness of the disaster risk and need for management strategies. • High cost of integrating disaster management strategies (Locals reluctance to purchase insurance, private insurance companies not willing to offer such policies. Individuals reluctant to retrofit buildings) For eg. Nepal (Political disagreements between parties slowed down rebuilding) For eg. Philippines (only 10 – 20% of buildings are insured)			

Name: _____ ()

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